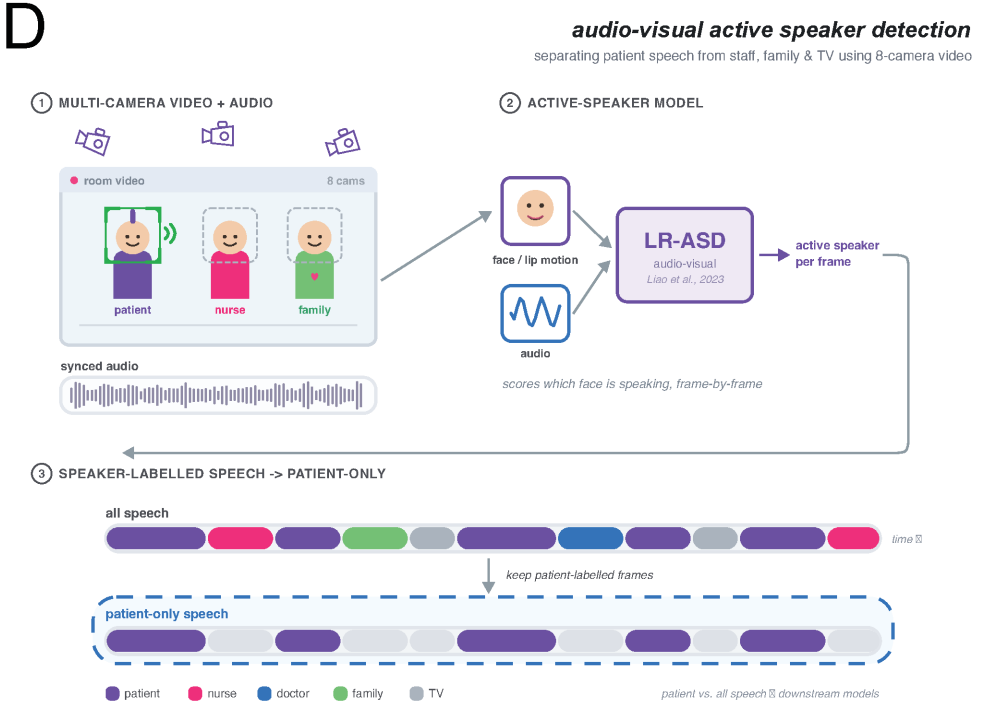
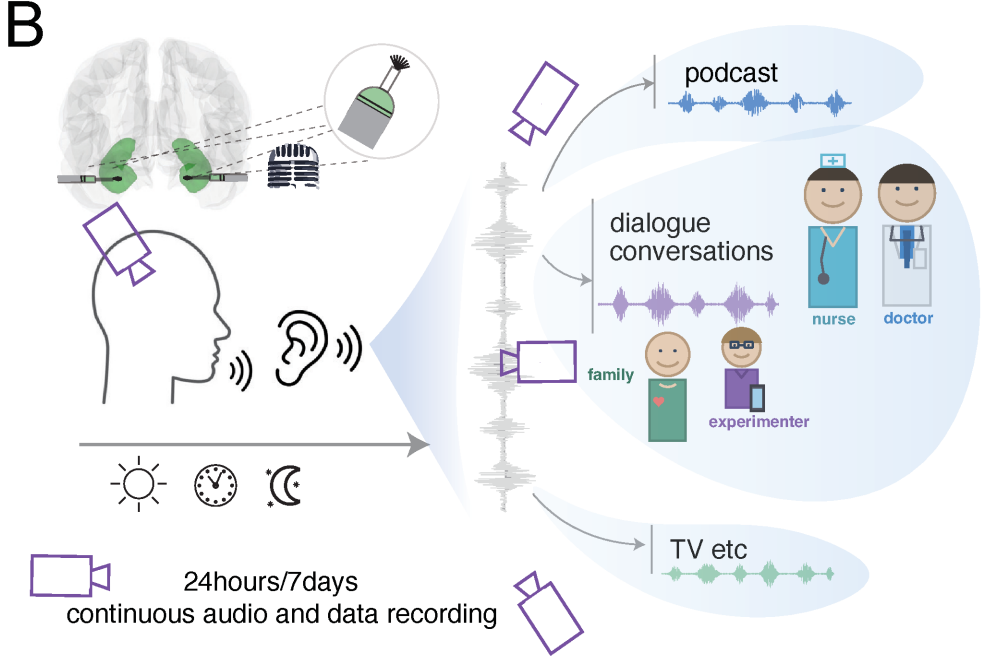
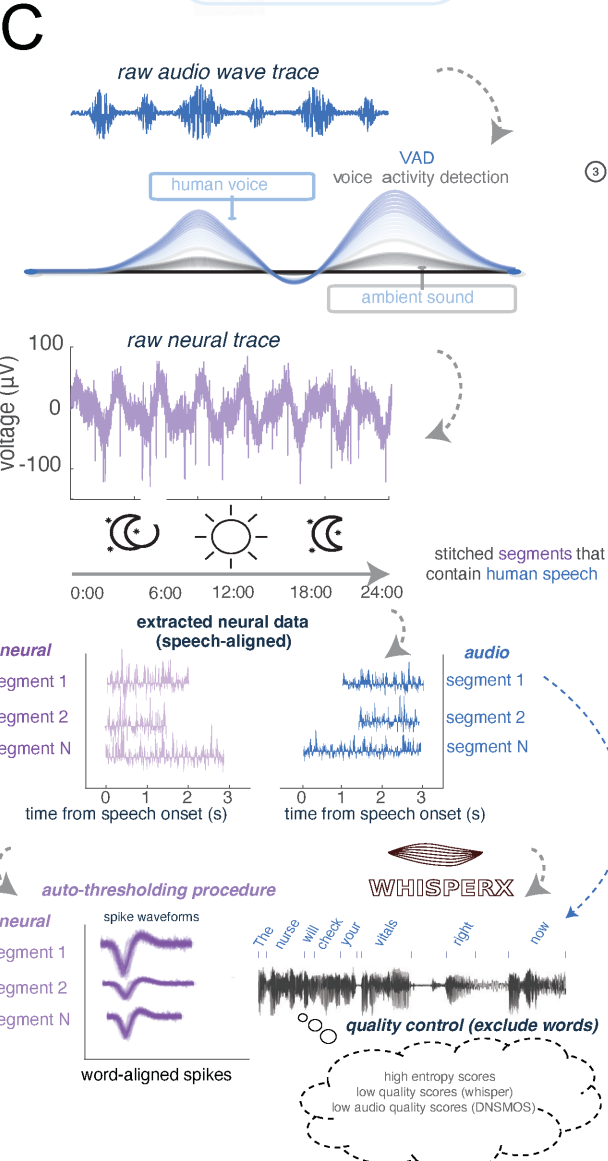
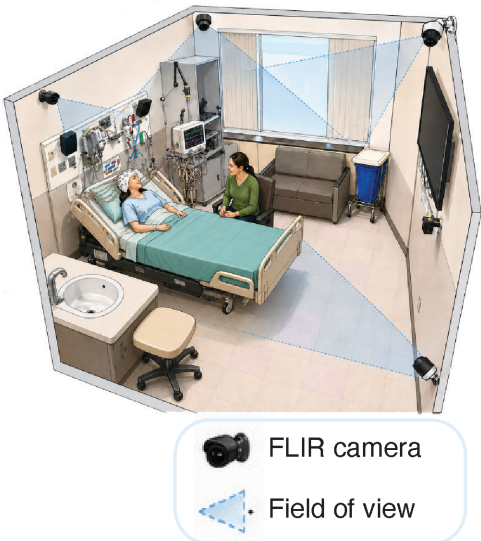




E

train encoding model

word embeddings

model weights

neuronal activity

Poisson GLM

$$\begin{bmatrix} \text{word}_1 \\ \text{word}_2 \\ \text{word}_3 \\ \dots \\ \text{word}_n \end{bmatrix} \begin{bmatrix} \text{dim}_1 & \text{dim}_2 & \text{dim}_3 & \dots & \text{dim}_{1000} \\ 0.19 & -0.46 & 2.56 & \dots & 2.69 \\ 3.35 & 3.38 & 0.25 & \dots & 0.01 \\ -2.53 & 0.46 & 0.08 & \dots & -2.33 \\ \dots & \dots & \dots & \dots & \dots \\ -1.52 & 3.88 & -0.61 & \dots & 2.79 \end{bmatrix} \times \begin{bmatrix} \theta_1 \\ \theta_2 \\ \theta_3 \\ \vdots \\ \theta_n \end{bmatrix} = \begin{bmatrix} N_1 \\ N_2 \\ N_3 \\ \vdots \\ N_b \end{bmatrix}$$

